# Annotations in RED
# Prototype Concept

## 1 Summary

We are prototyping a scale model of a building with the roof, shutter, and shutter positioning all the same as the final design. We will be testing the mechatronic function and control, the ability to cool the building's interior.

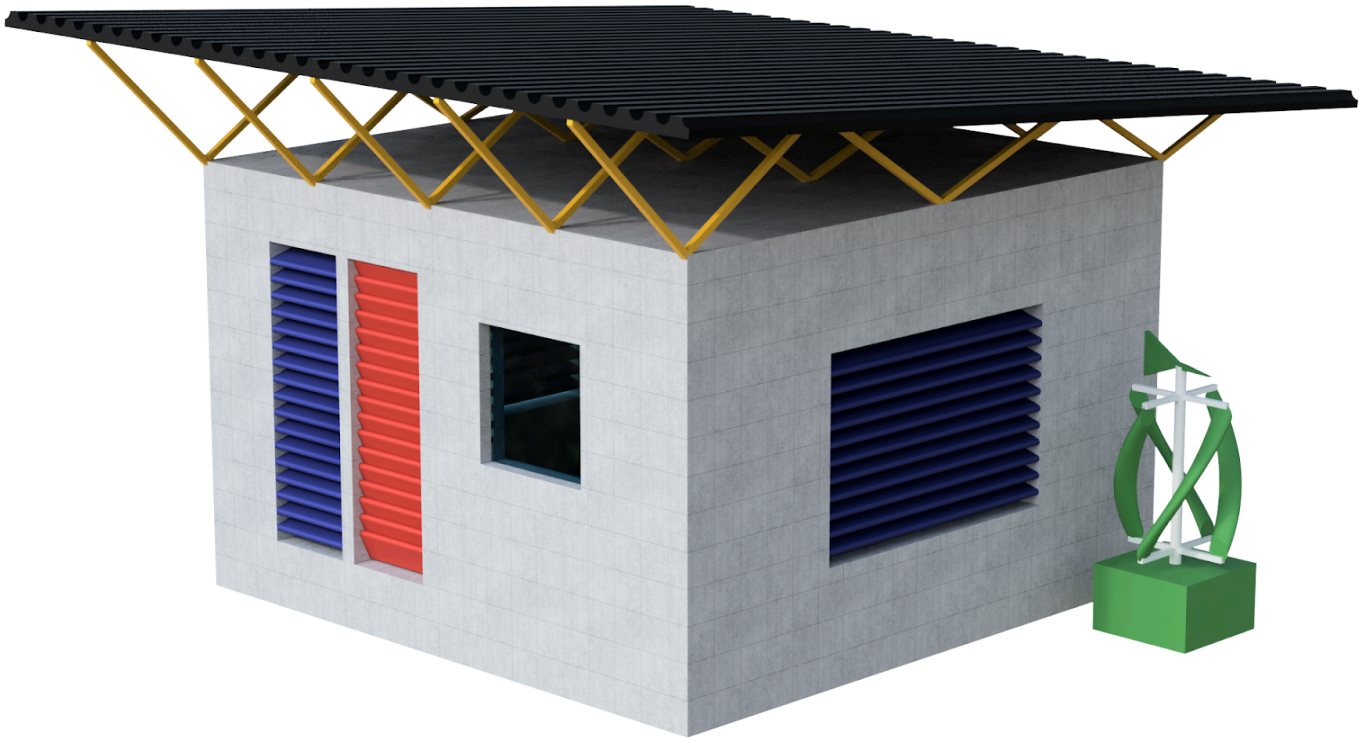

Figure 1: CAD model of our proposed design. We are getting rid of the stand alone power generation.

There are 4 subsystems in our prototype:

- We are modifying the defined "subsystems" to fit more with the definitions in the phase 3 rubric, as well as to make each subsystem modular and easier to test individually, and to have somewhat easier integration.
- The structural subsystem, which consists of ~~the scale house with the roof and shutter positioning.~~ Just the roof. We decided not to count the house as part of any subsystem as we did not have any requirements for it.
- The ~~mechatronic subsystem~~ window actuation subsystem (renaming for clarity), which consists of the ~~shutter blades~~ slats (renaming for clarity) and the motors controlling them.
- The sensor subsystem, which consists of two temperature sensors, one inside and one outside of the building, a wind speed sensor (anemometer), and a wind direction sensor (clarified that these are two distinct things). ~~and a wind speed and direction sensor.~~
- The control subsystem, which consists of the code used to optimize cooling ~~the wind flow.~~

We will not be prototyping the power supply, nor the material choice used in the final design. The former would rely on us having access to the Ghana power grid (or a suitable substitute) so we could design the necessary electronics. The latter would require us to perform complicated scaling calculations or to create a full-sized model of the house; the former is prone to error while the latter is completely out of our budget, ~~although computational methods could be used to select a good material in a different prototype.~~ We did not do this.

We will test the airflow. While we cannot prototype temperature control physically, as we do not have the means to generate a temperature-controlled flow of air into the model, nor can we ensure that it scales properly to the size of a house, we still want to test if the air does effectively flow into the house as desired. ~~This will be done through simulations, such as SimScale's Indoor Air Simulation [1].~~ After consultation with the teaching team, we got permission to use a smoke machine to track airflow in our fully built design.

Custom-built anemometers for wind speed and direction will be prototyped. These sensors are relatively small and build details are readily available, so can be prototyped at ~~full scale [2].~~ The anemometer is at a reasonable scale, but we decided not to prototype the wind direction sensor to "full scale", as that would have used a lot of unnecessary resources, and would have been very big.

We are also prototyping the mechatronic motor-control aspect of our design since dust is an issue in northern Ghana [3]. ~~Given we can't simply hand-calculate how dust affects our mechatronic components, using the prototype to test the dust resistance will be useful.~~ After consulting with the teaching team, we decided not to test dust resistance, as that isn't really possible. It also allows us to have a proof-of-concept of our motor control algorithm for the shutters.

We also decided to prototype the temperature sensor modules. We aren't sure how the placement of temperature sensors will affect the readings. We will use the prototype to test this, and see if we will need to add additional sensors to get an accurate measurement of the interior and exterior temperatures.

**2 Fabrication Plan**

MyFab has all the components we need to develop our physical prototypes. Several ~~DHT11~~ changed to AM2320 sensors to have easier circuitpython support temperature and humidity sensors can be purchased, and the data can be read over serial via the Pico's GPIO Pins [4]. ~~Additionally, to actuate our design, we can rent multiple 6V 100 RPM motors and an L298N motor driver, along with a 9V battery connector and a buck converter to power the electrical subsystem.~~ After consultation with MyFab staff (meeting with Sarah), we decided to instead use a servo motor, as those could be powered by 4xAA batteries and did not need a motor driver, significantly simplifying our design, aligning with the goals set in our team charter. ~~In the future, we may consider using an Arduino instead of a Pico, as there are more components compatible with the Arduino's 5V logic.~~ The pico has a 5V output, we learned that this is pretty nonsense.

~~Further, our CAD model is more or less complete and ready for air flow simulations. We can begin these immediately and experiment with different window shapes and sizes for optimal flow. These would validate our design. As well, we intend to show these simulations to our professors and TAs for feedback on any fluid mechanics considerations we missed, as this is out of our area of expertise.~~ As previously mentioned, we preferred physical testing to simulations, and we eventually (after a long, long series of emails) got permission from EHS to do smoke testing.

## 3 Conclusion

Our design aims to assist home cooling in Northern Ghanaian communities. Our prototypes test the mechatronic system, the cooling functionality, and temperature, wind, and humidity sensors. ~~Cooling is difficult to test at a reasonable scale, and given our lack of testing equipment, we opt to simulate airflow and temperatures based on our CAD model.~~

## Works Cited

[1] "Cloud-Native Indoor Environment Simulation | SimScale," *SimScale*, Dec. 17, 2025. https://www.simscale.com/indoor-environment-simulation

[2] "A DIY Wind Gauge (Anemometer) based on ESP32." https://smartsolutions4home.com/ss4h-wg-wind-gauge/

[3] "Harmattan intensification," *GMet CMS*. https://www.meteo.gov.gh/alerts/harmattan-intensification/

[4] "DHT11–Temperature and humidity sensor," *Components101*. https://components101.com/sensors/dht11-temperature-sensor